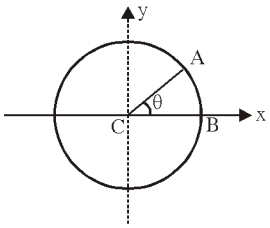
SECTION-I

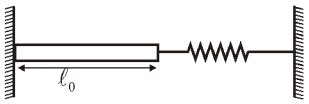
Single Correct Answer Type **12 Q. [3 M (–1)]**

1. Two lines AC and BC are drawn on a circular disc as shown in figure. The disc is made of a material with linear expansion coefficients $\alpha_x > \alpha_y$. Here α_x and α_y are linear expansion coefficient for expansion in x-direction and y-direction respectively. C is the centre of disc. If the disc is heated uniformly. Angle ACB (=θ) will



(A) increase
(B) decrease
(C) remains constant
(D) may increase or decreases depending on value of temperature increment

2. A rod of mass M, area of cross section A and length ℓ_0 is connected with a spring as shown in figure. If coefficient of linear expansion of rod is α and initially no extension was there in the spring of spring constant k, then the stress developed in rod when its temperature is increased by ΔT, is
[Young's Modulus of material of the rod is Y]



(A) $\frac{k[Y\alpha\Delta T]\ell_0}{[k\ell_o + YA]}$ (B) $Y\alpha\Delta T$ (C) $\left[\frac{k\ell_o + YA}{k\ell_o}\right]Y\alpha\Delta T$ (D) $\left[\frac{k\ell_o + Y}{k\ell_o}\right]Y\alpha\Delta T$

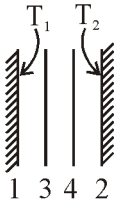
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3. A wire of cross-section A is stretched horizontally between two clamps located $2l$ meters apart. A weight W kg is suspended from the mid point of the wire. If the mid point sags vertically through a distance $x << l$ the strain produced is :-

(A) $\frac{2x^2}{l^2}$ (B) $\frac{x^2}{l^2}$ (C) $\frac{x^2}{2l^2}$ (D) none of these

4. In an industrial process 10 kg of water per hour is to be heated from 20°C to 80°C . To do this steam at 150°C is passed from a boiler into a copper coil immersed in water. The steam condenses in the coil and is returned to the boiler as water at 90°C. How many kg of steam is required per hour.
(Specific heat of steam = 1 cal/g°C, Latent heat of vaporisation = 540 cal/g)
(A) 1g (B) 1 kg (C) 10 g (D) 10 kg

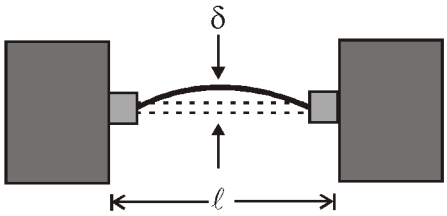
5. Two large black plane surfaces are maintained at constant temperature T_1 and $T_2(T_1 > T_2)$. Two thin black plates are placed between the two surfaces and in parallel to these. After some time, steady conditions are obtained. What is the ratio of heat transfer rate between plate-1 & plate-3 to the ratio of original (when plate-3 & plate-4 was not present) heat tranfer rate between plate-1 & plate-2 (η) in steady state?



(A) $\eta = \frac{1}{2}$ (B) $\eta = \frac{1}{3}$ (C) $\eta = 1$ (D) $\eta = 0$

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6. A long, thin metal bar of length ℓ is clamped rigidly at its ends at temperature t_0. When the temperature is increased to t, the expanding bar will bow out, as shown below. If the bowing is not too large, a fair first approximation to the shape of the bar is two equal straight segments in the form of a wide V. What is the arch δ of the bow as a function of t? (This is the distance between the corner of the V and the straight line that represents the form of the bar at t_0.)

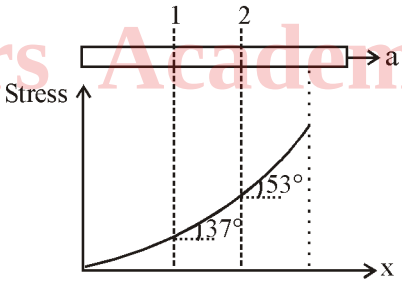


(A) $\delta = (\ell/2)\sqrt{2\alpha(t-t_0)}$ (B) $\delta = (\ell)\sqrt{\alpha(t-t_0)}$

(C) $\delta = (\ell/2)\sqrt{\alpha(t-t_0)}$ (D) $\delta = (\ell)\sqrt{2\alpha(t-t_0)}$

7. Rod of constant cross-section moves towards right with constant acceleration. Graph of stress and distance from left end is given as in figure. If density of material of rod at cross section 1 is 9 gm/cm^3. Find density at cross section 2.

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(A) 16 gm/cm^3 (B) 20 gm/cm^3 (C) 24 gm/cm^3 (D) 12 gm/cm^3

8. 'Gulab Jamuns' (assumed to be spherical) are to be heated in an oven. They are available in two sizes, one twice bigger (in radius) than the other. Pizzas (assumed to be discs) are also to be heated in oven. They are also in two sizes, one twice big (in radius) than the other. All four are put together to be heated to oven temperature. Choose the correct option from the following:

(A) Both size gulab jamuns will get heated in the same time.

(B) Smaller gulab jamuns are heated before bigger ones.

(C) Smaller pizzas are heated before bigger ones.

(D) Bigger pizzas are heated before smaller ones.

9. Three very large plates of same area are kept parallel and close to each other. They are considered as ideal black surfaces and have very high thermal conductivity. The first and third plates are maintained at temperatures 2T and 3T respectively. The temperature of the middle (i.e. second) plate under steady state condition is **[JEE 2012]**

(A) $\left(\frac{65}{2}\right)^{1/4} T$ (B) $\left(\frac{97}{4}\right)^{1/4} T$ (C) $\left(\frac{97}{2}\right)^{1/4} T$ (D) $(97)^{1/4} T$

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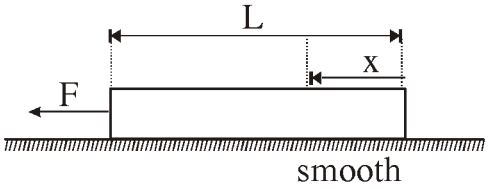
10. There is formation of layer of ice x cm thick on water, when the temperature of air is $-\theta^\circ C$ (less than freezing point). The thickness of layer increases from x to y in the time t, then the value of t is given by-

(A) $\frac{(x+y)(x-y)\rho L}{2K\theta}$ (B) $\frac{(x-y)\rho L}{2K\theta}$ (C) $\frac{(x+y)(x-y)\rho L}{K\theta}$ (D) $\frac{(x-y)\rho LK}{2\theta}$

11. Two thin walled spheres of different materials, one with double the radius and one-fourth wall thickness of the other, are filled with ice. If the time taken for complete melting of ice in the sphere of larger radius is 25 minutes and that for smaller one is 16 minutes, the ratio of thermal conductivities of the materials of larger sphere to the smaller sphere is

(A) 4 : 5 (B) 25 : 1 (C) 1 : 25 (D) 8 : 25

12. A rod of mass m, uniform cross sectional area A and length L is accelerated by applying force F as shown in figure on a smooth surface. If Young's modulus of elasticity of the material of rod is Y. (Consider x as measured from the right end)



(A) Tension in rod as a function of distance x is $\frac{Fx}{2L}$.

(B) Strain in rod is $\frac{F}{2AY}$.

(C) Elastic potential energy stored in the rod is $\frac{F^2L}{8AY}$.

(D) There is no stress in rod.

Multiple Correct Answer Type 6 Q. [4 M (–1)]

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13. A metal cylinder of very high conductivity and mass 0.5 kg is heated electrically by a 12 W heater in a room at 15°C. The cylinder temperature rises to 25°C in 5 min and finally becomes constant at 45°C. Assuming that the rate of heat loss is proportional to the excess temperature over the surroundings

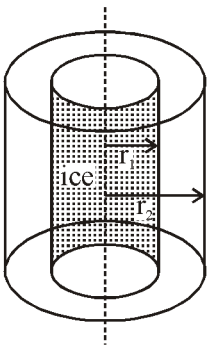
(A) The rate of loss of heat of the cylinder to surrounding at 20°C is 2 W

(B) The rate of loss of heat of the cylinder to surrounding at 45°C is 2 W

(C) Specific heat capacity of metal is $\frac{240}{\ell n(3/2)}$ J/kg°C

(D) None of these

14. A 100 cm long cylindrical flask with inner and outer diameter 2 cm and 4 cm respectively is completely filled with ice as shown in the figure. The constant temperature outside the flask is 40°C. (Thermal conductivity of the flask is 0.693 W/m°C, L_{ice} = 80 cal/gm & $\ell n2 = 0.693$).



(A) Rate of heat flow from outside to the flask is 80 π J/s.

(B) The rate at which ice melts is $\frac{\pi}{4200}$ kg/s

(C) The rate at which ice melts is 100 π kg/s

(D) Rate of heat flow from outside to flask is 40 π J/s

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15. Which of the following statements are true?

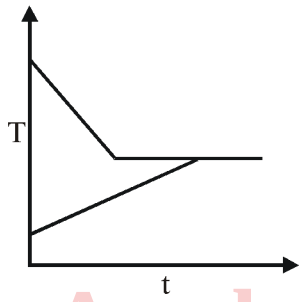
(A) Hole in the wall of a cavity radiator behaves like a black body.

(B) Hole in the wall of a cavity radiator does not act like a black body.

(C) When a body is kept in a surrounding of low temperature it does not absorb any energy from the surroundings.

(D) When a body is kept in surrounding of low temperature it simultaneously radiates heat to the surroundings and absorbs heat from the surroundings.

16. A sample A of liquid water and a sample B of ice of identical mass are kept in two neighbouring chambers in an otherwise insulated container. The chambers can exchange heat with each other. The graph of temperatures of the two chambers is plotted with time. $S_{ice} = \frac{S_{water}}{2}$:-



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(A) Finally the contents in sample A is water.

(B) Equilibrium temperature is freezing point of water

(C) Ice melts partly

(D) Finally the contents in sample B is ice only.

17. 50 gm ice at –10°C is mixed with 10gm steam at 100°C. When the mixture finally reaches its steady state inside a calorimeter of water equivalent 1.5 gm then : [Assume calorimeter was initially at 0°C, Take latent heat of vaporization of water = 540 cal/gm, Latent heat of fusion of water = 80 cal/gm and specific heat capacity of water = 1 cal/gm-°C, specific heat of ice = 0.5 cal/gm°C]

(A) Mass of water remaining is 60 gm (B) Mass of ice remaining is 3 gm

(C) Mass of steam remaining is 1.20 gm (D) Final temperature is between 0°C and 100°C

18. The temperature of earth is maintained by a dynamic equilibrium between Sun and Earth. Sun & Earth can be assumed to be black bodies :

(A) If the power output of sun would double with changing the temperature, equilibrium temperature of earth also doubles.

(B) If the radius of sun doubles without changing its' power, its surface temperature would decrease by factor of $\sqrt{2}$.

(C) If the radius of earth doubles without any change in sun, it's equilibrium temperature would increase by factor of $\sqrt{2}$.

(D) If the distance between earth and sun would decrease by a factor of 2, the equilibrium temperature of earth would increase by factor of $\sqrt{2}$.

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Linked Comprehension Type (1 Para × 3Q.) [3 M (-1)]
(Single Correct Answer Type)

Paragraph for Question No. 19 to 21

Multi-layer insulation, or MLI, is thermal insulation composed of multiple layers of thin sheets often used on spacecraft. It is one of the main items of the spacecraft thermal design, primarily intended to reduce heat loss by thermal radiation. In its basic form, it does not appreciably insulate against other thermal losses such as heat conduction or convection. It is therefore commonly used on satellites and other applications in vacuum where conduction and convection are much less significant and radiation dominates. MLI gives many satellites and other space probes the appearance of being covered with gold foil.

The principle behind MLI is radiation balance. To see why it works, start with a concrete example - imagine a square meter of a surface in outer space, at 300 K, with an emissivity of 1, facing away from the sun or other heat sources. From the Stefan-Boltzmann law, this surface will radiate 460 watts. Now imagine we place a thin (but opaque) layer 1 cm away from the plate, thermally insulated from it, and also with an emissivity of 1. This new layer will cool until it is radiating 230 watts from each side, at which point everything is in balance. The new layer receives 460 watts from the original plate. 230watts is radiated back to the original plate, and 230 watts to space. The original surface still radiates 460 watts, but gets 230 back from the new layers, for a net loss of 230 watts. So overall, the radiation losses have been reduced by half by adding the additional layer.

More layers can be added to reduce the loss further. The blanket can be further improved by making the outside surfaces highly reflective to thermal radiation, which reduces both absorption and emission. The performance of a layer stack can be quantified in terms of its overall heat transfer coefficient U, which defines the radiative heat flow rate Q between two parallel surfaces with a temperature difference ΔT and area A as

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$$Q = UA\Delta T$$

Theoretically, the heat transfer coefficient between two layers with emissivities $\in_1$ and $\in_2$, under vacuum, is

$$U = 4\sigma T^3 \frac{1}{\frac{1}{\in_1}+\frac{1}{\in_2}-1},$$

where T is the mean of the temperatures (in K) of the two layers and $\sigma = 5.7\times10^{-8}\,Wm^{-2}K^{-4}$ is the Stefan-Boltzmann Constant. If each layer has the same emissivity $\in$ on both sides, then a stack of N layers placed between two high-emissivity surfaces will have an overall heat transfer coefficient

$$U = 4\sigma T^3 \frac{1}{N\left(\frac{2}{\in}-1\right)+1}$$

Clearly, increasing the number of layers and decreasing the emissivity, lower the heat transfer coefficient, which is equivalent to a higher insulation value. In this equation, it is assumed that the temperature difference is small compared to the absolute temperature.

19. Main purpose of multilayer insulation is insulation against heat transfer, where heat transfer is mainly due to :-

(A) Conduction
(B) Conduction & radiation
(C) Conduction & convection
(D) Radiation

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20. In space there is MLI consisting of 50 layers. Each layer has emissivity ($\in = 0.2$) both sides & outer most temperature is 550 K and inner most temperature is 100 K less than outer most then find over all heat transfer coefficient (in $Wm^{-2}k^{-1}$) :-

(A) 6.33 (B) 6.33×10^{-2} (C) 12.66×10^{-3} (D) 6.33×10^{-3}

21. For a particular plate heat transfer due to conduction and radiation are equal. Width of a plate is 2mm, mean temperature is 1000 K and emissivity of inner and outer surface are 1 & $\frac{1}{2}$ respectively. Then find conductivity of plate material (in SI) :-

(A) 0.228 (B) 11 (C) 1.11 (D) Data is insufficient

SECTION-II

Numerical Answer Type Question **4 Q. [3(0)]**

(upto second decimal place)

22. A liquid of volumetric thermal expansion coefficient 200×10^{-6} /°C and bulk modulus = 1.2×10^9 Pa is filled in a spherical tank of negligible heat expansion coefficient. Its radius is 25 cm and wall thickness is 2 mm. When the temperature of the liquid is raised by 20 °C, find the tensile stress developed (in MPa) in the wall of the tank?

23. A rod of negligible heat capacity has length 40 cm, area of cross-section 1.0 cm^2 and thermal conductivity 100 W/m-°C. The temperature of one end is maintained at 0°C and that of the other end is slowly and linearly varied from 0°C to 60°C in 10 minutes. Assuming no loss of heat through the sides, find the total heat transmitted through the rod in these 10 minutes (in J).

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24. A cylindrical block of length 0.4 m an area of cross-section 0.04m^2 is placed coaxially on a thin metal disc of mass 0.4 kg and of the same cross-section. The upper face of the cylinder is maintained at a constant temperature of 400K and the initial temperature of the disc is 300K. If the thermal conductivity of the material of the cylinder is 10 watt/m-K and the specific heat of the material of the disc in 600 J/kg-K, how long will it take for the temperature of the disc to increase to 350K? Assume, for purposes of calculation, the thermal conductivity of the disc to be very high and the system to be thermally insulated except for the upper face of the cylinder.

25. An electric heater is used in a room of total wall area 50m^2 to maintain a constant temperature of 15°C inside when the outside temperature is –10°C. The walls have two different layers. The inner layer is of cement of thickness 2 cm, while the outer layer is of brick of thickness 10 cm. Assume that there is no loss of heat from the floor and the ceiling. Take the thermal conductivities of cement and brick to be 0.2 W/m°C and 0.5 W/m°C, respectively. If another identical heater is now switched on, along with the old heater, find the steady state temperature (in °C) inside the room, after a long time, if the outside temperature has now dropped to –15°C.

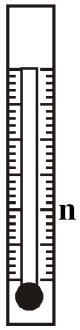
SECTION-III

Numerical Grid Type (Ranging from 0 to 9) **3 Q. [4 M (0)]**

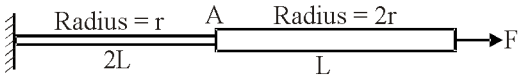
26. An object can be hanged to a 1.2m long wire of diameter 2 mm, and the wire does not break. However, if the object is slightly displaced and begins to slowly swing, the wire breaks. If another piece of wire having the same length and material but wider diameter is used, the speed of the same swinging object at lowest point can even be 6 m/s. Calculate the least diameter (in mm) of this wider wire.

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27. A thermometer has a spherical bulb of volume 1 cm^3 having 1 cm^3 of mercury. A long cylindrical capillary tube is connected to spherical bulb. Volumetric coefficient of expansion of mercury is $1.8 \times 10^{-4}\ K^{-1}$; cross-section area of capillary is $1.8 \times 10^{-4}\ cm^2$. Ignoring expansion of glass, how far apart (in cm) on the stem are marks indicating 1K temperature change.



28. Two steel wires of radii r and 2r are connected together end to end and tied to a wall as shown. The force stretches the combination by $\frac{27}{4}$ mm. How far does the junction point A move. (in mm)



Subjective Type **2 Q. [4 M (0)]**

29. There are two spheres of same radius and material at same temperature but one being solid while the other hollow. Which sphere will expand more if
(a) they are heated to the same temperature, (b) same heat is given to them ?

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30. A thin uniform metallic rod of length 0.5 m and radius 0.1 m rotates with an angular velocity 400 rad/s in a horizontal plane about a vertical axis passing through one of its ends. Calculate (a) tension in the rod and (b) the elongation of the rod. The density of material of the rod is 10^4 kg/m^3 and the Young's modulus is 2×10^{11} N/m^2.

GR # Elasticity, Thermal Expansion, Calorimetry & Heat Transfer — ANSWER KEY

SECTION-I

Single Correct Answer Type **12 Q. [3 M (–1)]**

1. Ans. (B)	**2. Ans. (A)**	**3. Ans. (C)**	**4. Ans. (B)**
5. Ans. (B)	**6. Ans. (A)**	**7. Ans. (A)**	**8. Ans. (B)**
9. Ans. (C)	**10. Ans. (A)**	**11. Ans. (D)**	**12. Ans. (B)**

Multiple Correct Answer Type **6 Q. [4 M (–1)]**

13. Ans. (A, C)	**14. Ans. (A,B)**	**15. Ans. (A,D)**	**16. Ans. (B, D)**
17. Ans. (A,D)	**18. Ans. (B,D)**		

Linked Comprehension Type (Single Correct Answer Type) **(1 Para × 3Q.) [3 M (-1)]**

19. Ans. (D)	**20. Ans. (B)**	**21. Ans. (A)**

SECTION-II

Numerical Answer Type Question (upto second decimal place) **4 Q. [3(0)]**

22. Ans. 300	**23. Ans. 450.00**	**24. Ans. 166.3**	**25. Ans. 35.00**

SECTION-III

Numerical Grid Type (Ranging from 0 to 9) **3 Q. [4 M (0)]**

26. Ans. 4	**27. Ans. 1**	**28. Ans. 6**

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